

Homework 7

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Copper Concentrations

```
copper <- read.csv("EPA.09.Ex.16.4.copper.txt")
```

Question a

```
summary(copper)
```

```
##      Month      Well      Well.type      Copper.ppb
##  Min.   :1.0   Length:18   Length:18   Min.    : 4.20
##  1st Qu.:2.0   Class :character Class :character 1st Qu.: 7.00
##  Median :3.5   Mode  :character Mode  :character Median : 9.90
##  Mean   :3.5
##  3rd Qu.:5.0
##  Max.   :6.0
##                               Mean   :10.38
##                               3rd Qu.:11.45
##                               Max.    :21.50
```

```
by(copper$Copper.ppb, copper$Well.type, summary)
```

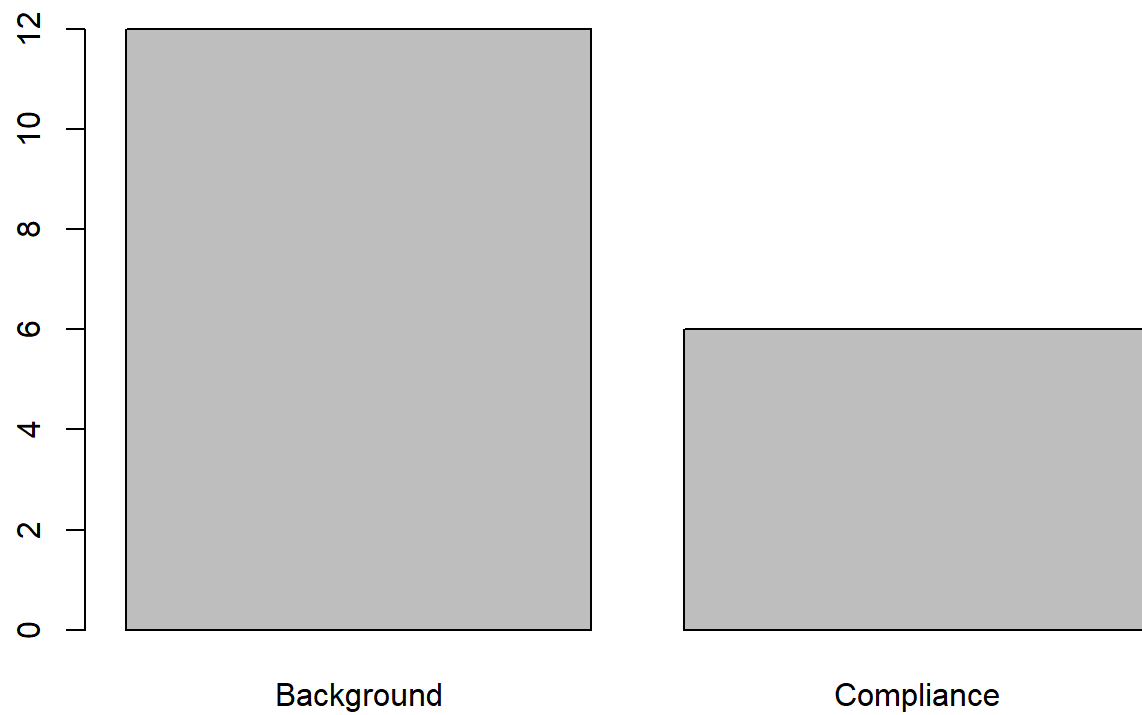
```
## copper$Well.type: Background
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      4.200  6.250   7.600   8.142 10.400  11.500
## -----
## copper$Well.type: Compliance
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      9.40  11.20   15.30   14.87  17.23   21.50
```

Question b

```
table(copper$Well.type)
```

```
##
## Background Compliance
##           12           6
```

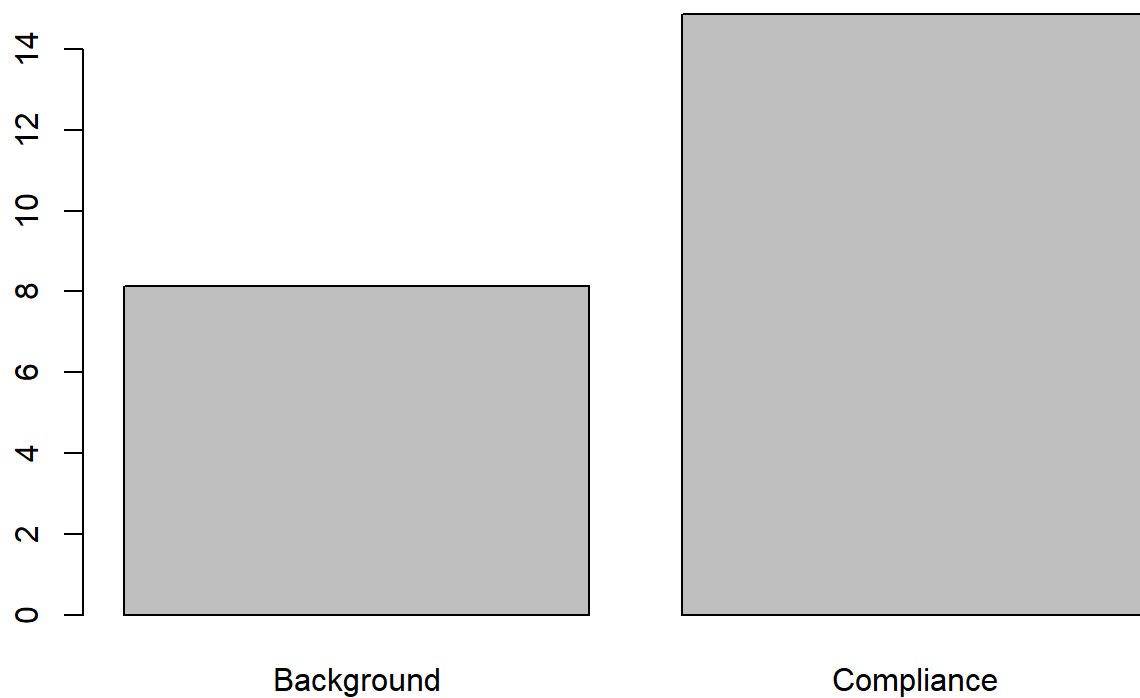
```
## give me a bar plot in base r
barplot(table(copper$Well.type))
```



```
avg_copp <- tapply(copper$Copper.ppb, copper$Well.type,  
                  mean)  
avg_copp
```

```
## Background Compliance  
##    8.141667 14.866667
```

```
barplot(avg_copp)
```



Question C

Null Hypothesis: Copper concentration is the same between compliance and background

Alternative Hypothesis: Copper concentration is more than that of background

```
background <- copper$Copper.ppb[copper$Well.type == "Background"]
compliance <- copper$Copper.ppb[copper$Well.type == "Compliance"]
background
```

```
## [1] 4.2 5.8 11.3 7.0 7.0 8.2 5.2 6.4 11.3 11.5 10.1 9.7
```

```
compliance
```

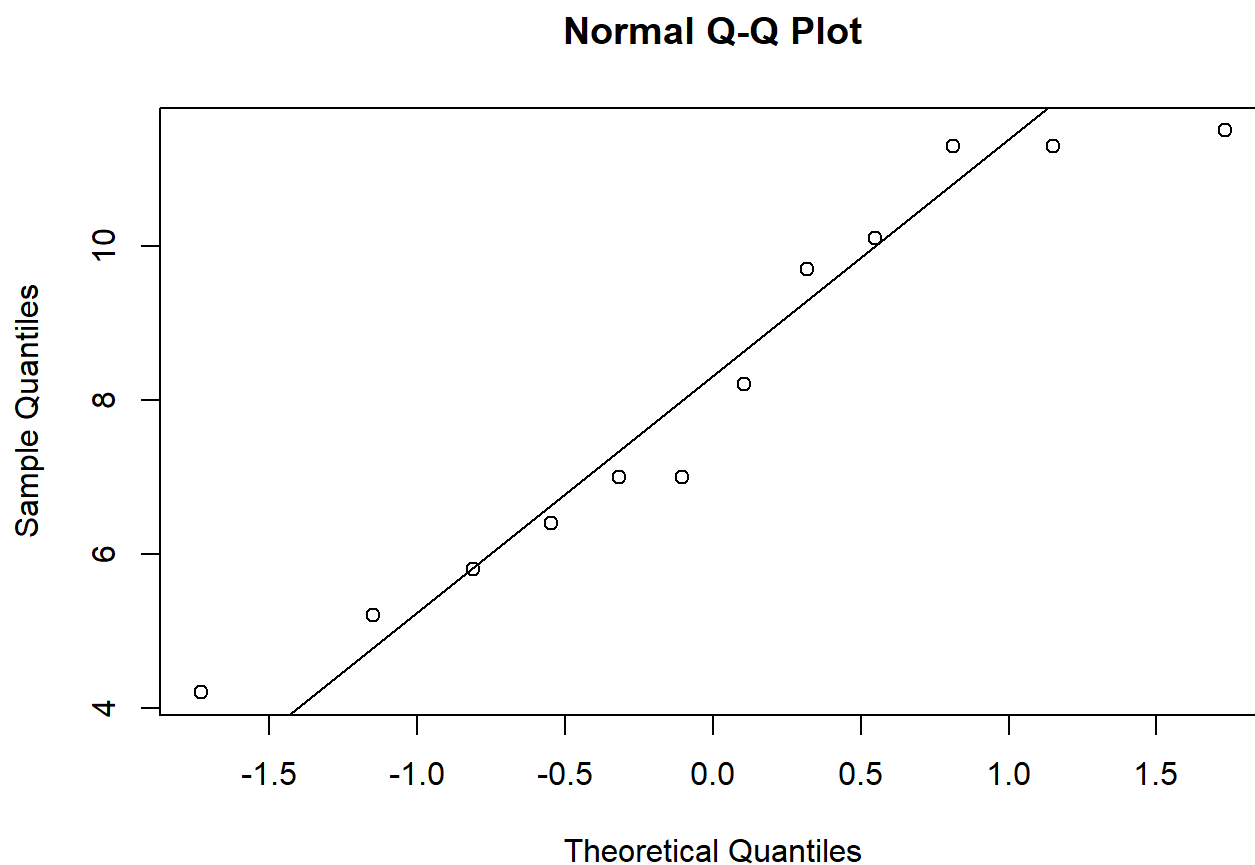
```
## [1] 9.4 10.1 14.5 16.1 21.5 17.6
```

```
t_result <- t.test(background, compliance,
                    alternative = "greater")
t_result
```

```
##  
## Welch Two Sample t-test  
##  
## data: background and compliance  
## t = -3.331, df = 6.6139, p-value = 0.9932  
## alternative hypothesis: true difference in means is greater than 0  
## 95 percent confidence interval:  
## -10.58387 Inf  
## sample estimates:  
## mean of x mean of y  
## 8.141667 14.866667
```

Question d

```
qqnorm(background)  
qqline(background)
```



Question e

```
well1 <- copper$Copper.ppb[copper$Well == "Well.1"]
well2 <- copper$Copper.ppb[copper$Well == "Well.2"]

pt <- t.test(well1, well2, paired = TRUE)
pt
```

```
##
## Paired t-test
##
## data: well1 and well2
## t = -2.5751, df = 5, p-value = 0.04973
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
## -3.563563075 -0.003103592
## sample estimates:
## mean difference
## -1.783333
```

Air quality measurements in New York

```
may_ozone <- airquality$Ozone[airquality$Month == 5]
aug_ozone <- airquality$Ozone[airquality$Month == 8]

t_ozone <- t.test(may_ozone, aug_ozone,
                  alternative = "two.sided")
t_ozone
```

```
##
## Welch Two Sample t-test
##
## data: may_ozone and aug_ozone
## t = -4.0749, df = 39.279, p-value = 0.0002169
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
## -54.38358 -18.30873
## sample estimates:
## mean of x mean of y
## 23.61538 59.96154
```

Lake Erie

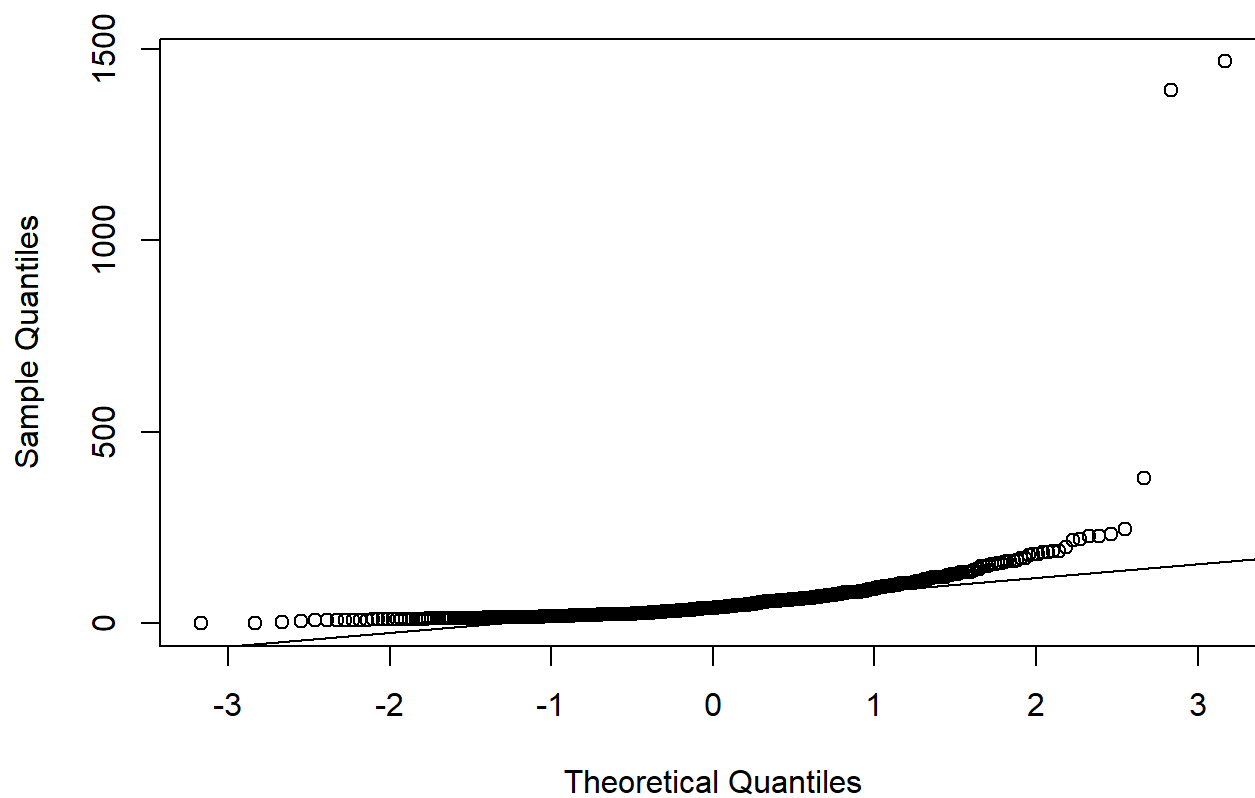
a

```
lake_erie_1 <- read.csv("LakeErie1.csv")  
#View(lake_erie_1)
```

```
tp <- lake_erie_1$TP  
chla <- lake_erie_1$chla  
  
# remove missing values  
tp <- na.omit(lake_erie_1$TP)  
chla <- na.omit(lake_erie_1$chla)
```

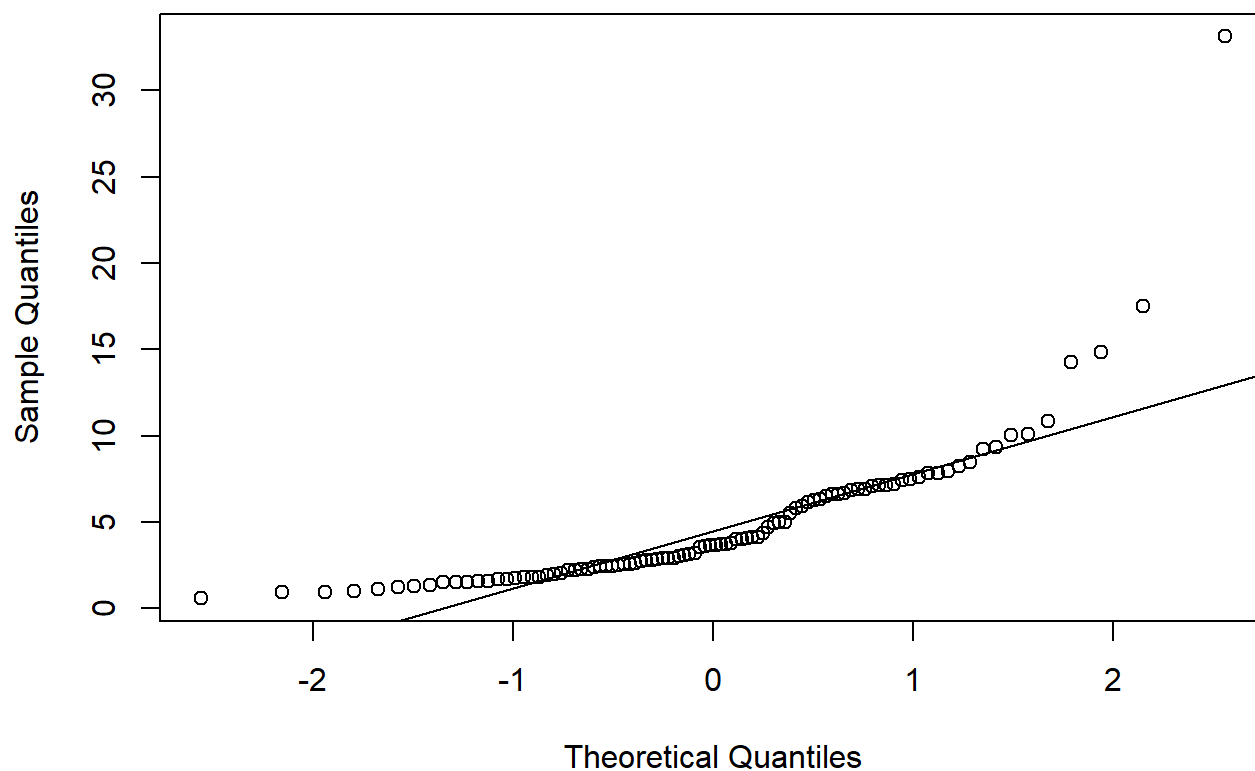
```
# for tp  
qqnorm(tp)  
qqline(tp)
```

Normal Q-Q Plot



```
qqnorm(chla)  
qqline(chla)
```

Normal Q-Q Plot



b

```
library(fitdistrplus)
```

```
## Warning: package 'fitdistrplus' was built under R version 4.5.3
```

```
## Loading required package: MASS
```

```
## Loading required package: survival
```

```
tp_dist <- fitdist(tp, "norm")
chla_dist <- fitdist(chla, "norm")
tp_dist
```

```
## Fitting of the distribution ' norm ' by maximum likelihood
## Parameters:
##      estimate Std. Error
## mean  58.06469    3.428629
## sd    87.61461    2.424403
```

```
chla_dist
```

```
## Fitting of the distribution ' norm ' by maximum likelihood
## Parameters:
##      estimate Std. Error
## mean 4.870938  0.4399708
## sd   4.310816  0.3111063
```

C

```
tp_mean <- tp_dist$estimate["mean"]
tp_mean
```

```
##      mean
## 58.06469
```

```
tp_sd <- tp_dist$estimate["sd"]
tp_sd
```

```
##      sd
## 87.61461
```

```
chla_mean <- chla_dist$estimate["mean"]
chla_sd <- chla_dist$estimate["sd"]
```

```
ks.test(tp, "pnorm", mean = tp_mean, sd = tp_sd)
```

```
## Warning in ks.test.default(tp, "pnorm", mean = tp_mean, sd = tp_sd): ties
## should not be present for the one-sample Kolmogorov-Smirnov test
```

```
##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data: tp
## D = 0.27765, p-value < 2.2e-16
## alternative hypothesis: two-sided
```

```
ks.test(chla, "pnorm", mean = chla_mean, sd = chla_sd)
```

```
## Warning in ks.test.default(chla, "pnorm", mean = chla_mean, sd = chla_sd): ties
## should not be present for the one-sample Kolmogorov-Smirnov test
```



```
##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  chl_a
## D = 0.16867, p-value = 0.008485
## alternative hypothesis: two-sided
```

d

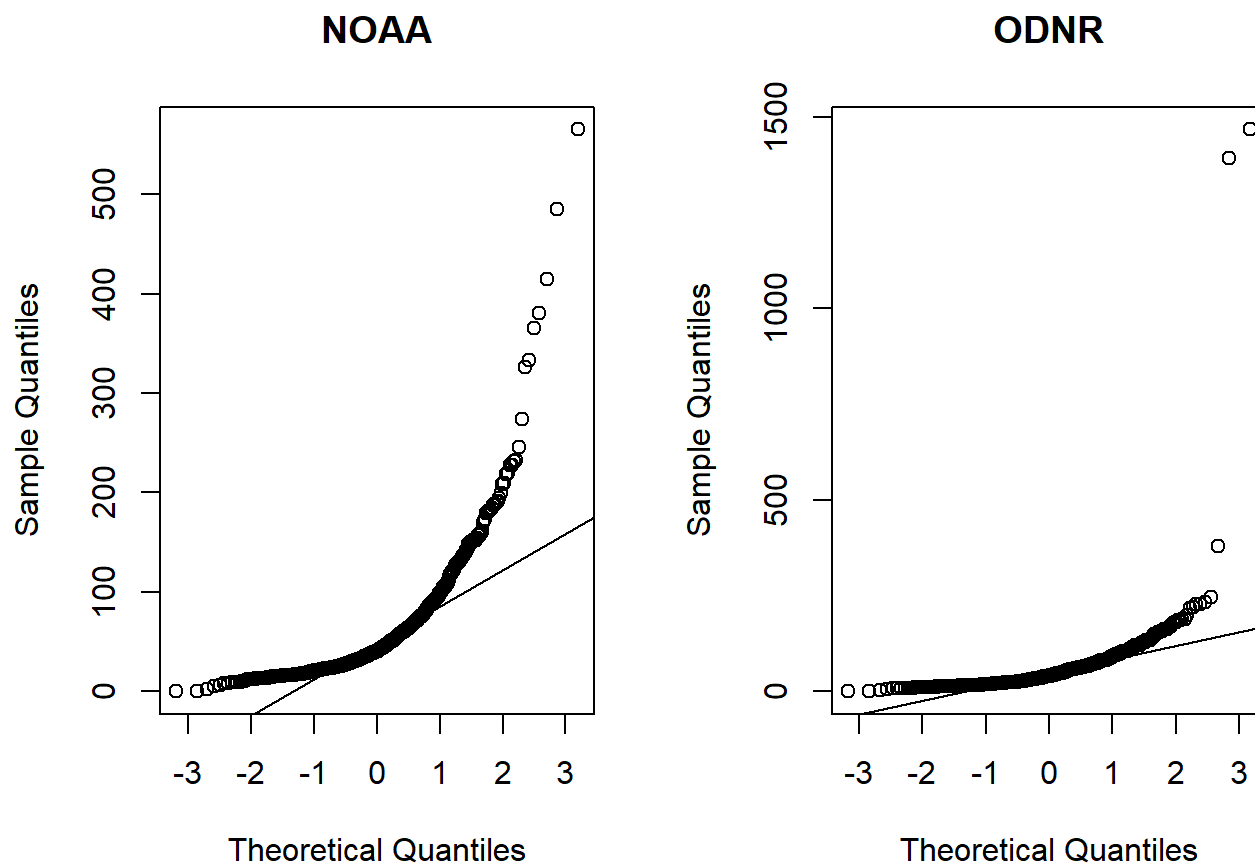
```
lake_erie_2 <- read.csv("LakeErie2.csv")
View(lake_erie_2)
```

```
tp_noaa <- na.omit(lake_erie_2$TP)
```

```
par(mfrow = c(1,2))

qqnorm(tp_noaa, main = "NOAA")
qqline(tp_noaa)

qqnorm(tp , main = "ODNR")
qqline(tp)
```



Uranium and total dissolved solids

```
hh <- read.csv("hh.csv")  
View(hh)  
  
u_conc <- hh$Uranium.ppb  
  
tds <- hh$TDS.mg.L  
  
cor(u_conc, tds)
```

```
## [1] 0.6374191
```